

assignment8 - DC circuits - fundamentals

 This is a preview of the published version of the quiz

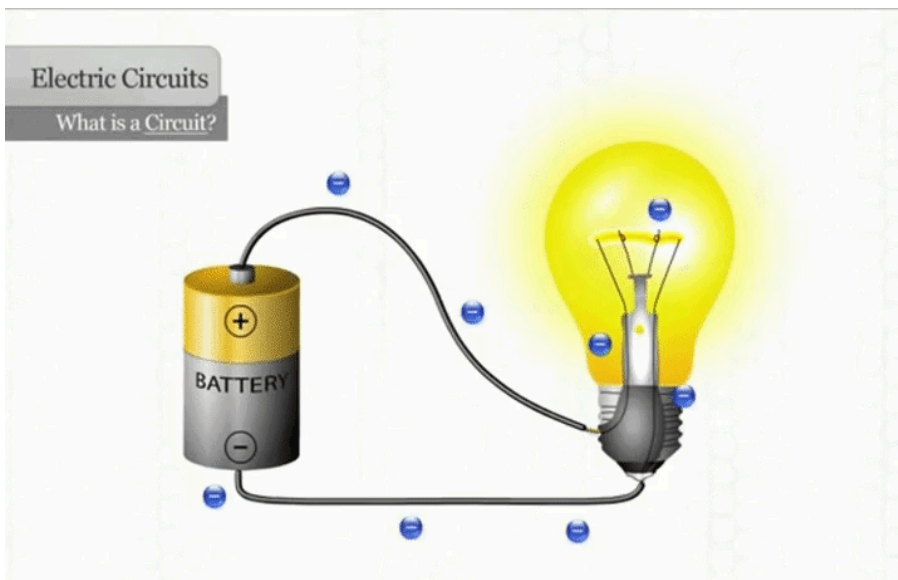
Started: Mar 22 at 6:54pm

Quiz Instructions

Make sure to try without looking at the hints (hide them). Remember to convert.

2 attempts. Last one counts .

Work with a group, in person. Collaborative learning is the best. Study textbook



Question 1 6.25 pts

Just as in hydraulic circuits there is water pressure, in electric circuits there is

See slides



voltage.



resistance.



current.



Question 2 6.25 pts

An ampere is a unit of electric



voltage.



resistance.



current.



Question 3 6.25 pts

Power is defined as the energy expended per unit of time. When translated to electrical terms, power is equal to



current divided by time.



current multiplied by voltage.



current multiplied by resistance.



voltage divided by time.



none of the above



Question 4 6.25 pts

What current is flowing in a wire if 0.67 C of charge pass a point in the wire in 0.30 s ?

hint: see slides for definition of current and equation



2.2 A



0.20 A



0.30 A



0.67 A



Question 5 6.25 pts

A charge of 12 C passes through an electroplating apparatus in 2.0 min . What is the average current in the apparatus?

hint: convert min to seconds.



0.60 A

☐

0.10 A

☐

6.0 A

☐

1.0 A



Question 6 6.25 pts

In units of measurement, power in watts is equal to

See my slides if you are not sure.

☐

amperes x ohms.

☐

none of the above

☐

volts/second.

☐

amperes/second.

☐

amperes x volts.



Question 7 6.25 pts

Which statement is correct?

☐

charge flows in a circuit

☐

resistance flows in a circuit

☐

voltage flows in a circuit

☐

all of the above



Question 8 6.25 pts

The unit of electrical resistance is the

☐

none of the above

☐

ohm.

☐

ampere.



volt.



Question 9 6.25 pts

One joule per coulomb is a unit of



voltage.



power.



current.



resistance.



energy.



Question 10 6.25 pts

A coulomb of charge flowing in a bulb filament powered by a 6-volt battery is provided with



6 joules.



6 amperes.



6 newtons.



6 watts.



6 ohms.



Question 11 6.25 pts

How much charge must pass by a point in a wire in 10 s for the current in the wire to be 0.50

A?



20 C



5.0 C



0.050 C



2.0 C



Question 12 6 pts

TO make a potato battery, in addition to the potato, load and connecting wire, you could use

☐

2 brass nails

☐

2 plastic rulers

☐

1 copper wire and 1 brass nail

☐

2 copper wires

☐

Question 13 4 pts

A positive ion is formed when (for sure)

☐

It depends on the element

☐

a neutral atom loses an electron

☐

a negative ion loses an electron

☐

a neutral atom gains an electron

☐

Question 14 4 pts

A battery supplies _____ - and wall outlet supplies _____

☐

AC current, DC current

☐

DC current, DC current

☐

DC current, AC current

☐

It depends on the voltage

☐

Question 15 6 pts

In class, I showed you that the voltage of the wall outlet was about:

☐

12V

☐

9V

☐

120V



1.5V



Question 16 6 pts

2 AA batteries are connected in series. The voltage is now

hint: see the slides for AA batteries



3V



1.5V



9V



6V



Question 17 6 pts

In a DC circuit (as shown in class) with wires, power supply and loads

charge carriers are the particles that "flow" in a circuit.



protons are the charge carrier



electrons are the charge carriers



ions are the charge carriers



frogs are the charge carriers

Not saved

Submit Quiz